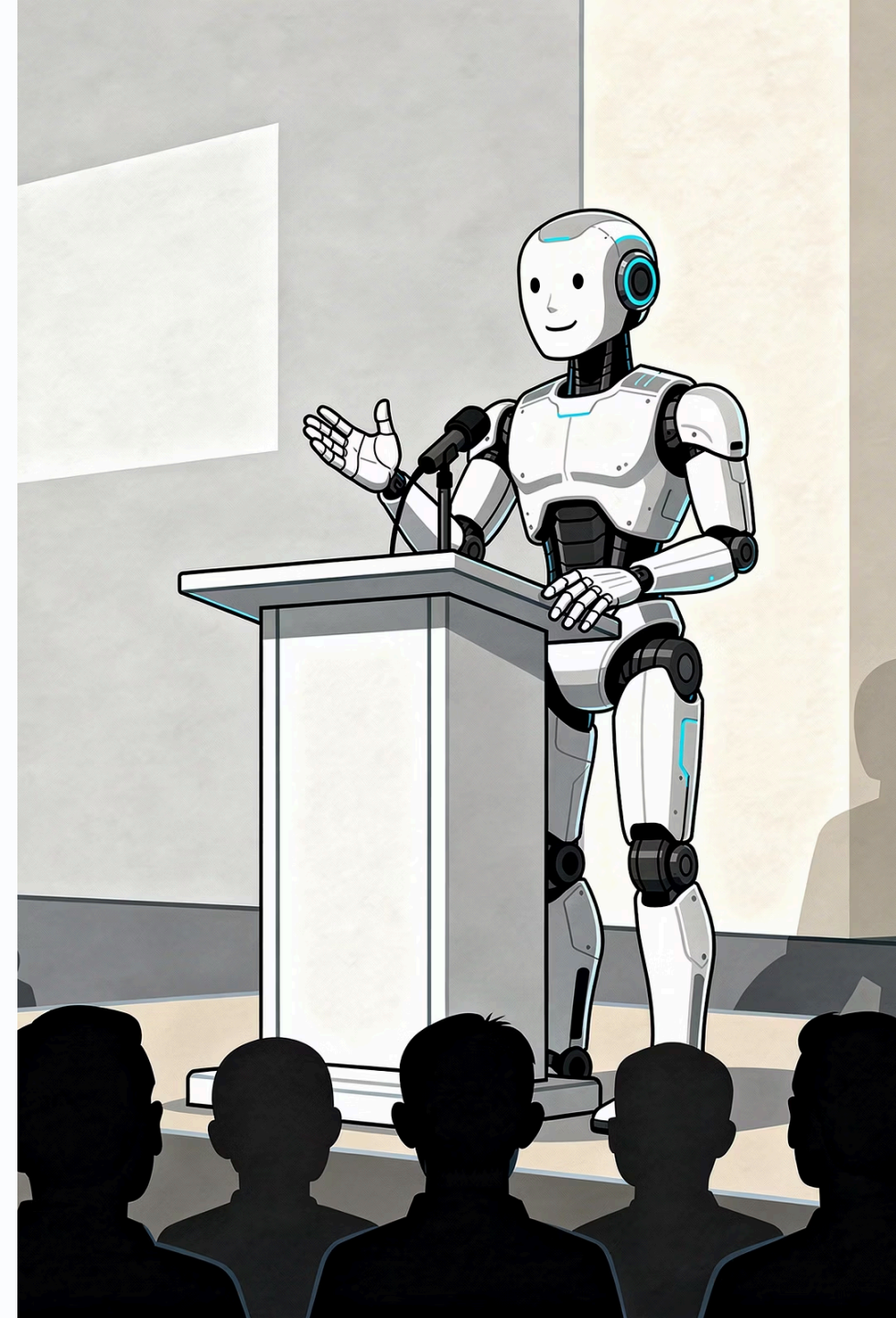


Session 9: AI Hallucinations - When AI Is Confidently Wrong

Last time you learned how LLMs predict words one at a time. Today, we look at what happens when those predictions go **wrong** - but the AI doesn't realize it.

⚠️ This is one of the most important sessions in the whole course. Pay close attention. 🧠



Objectives:

By the end of this session, you'll be able to:

01

Define It

Say what an **AI Hallucination** is

03

Understand Why

Know **WHY** AI makes stuff up

05

Catch It

Catch AI when it's making things up

02

Confidence vs. Accuracy

Tell the difference between **confidence** and **accuracy**

04

Real Stories

Spot real stories where AI got people in trouble

06

Use NotebookLM

Use an AI that's designed to hallucinate **LESS**



Recap & Homework Share

Welcome back! Let's see your LLM Detective work.

Compare the Tools

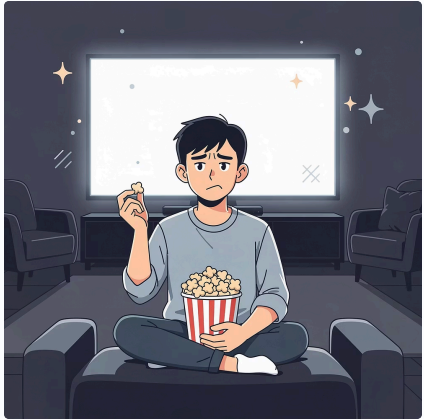
Which was more interesting - ChatGPT or Gemini? Did either of them say anything that seemed **wrong**?

Quick Check from Last Week

What's a Language Model? Does AI actually understand what it says? (*NO! It's a pattern parrot!*) What's a token?

Warm-Up - Confidently Wrong

Quick story: Have you ever been **100% sure** about something... and turned out to be totally wrong?



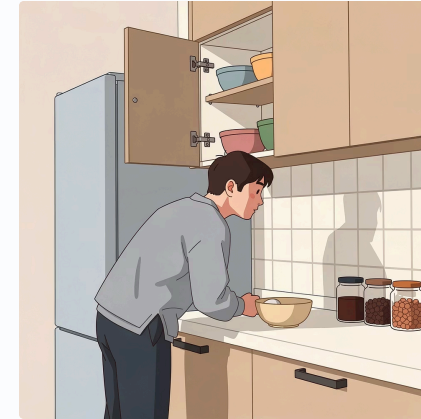
Movie Night

"This movie is going to be SO good!" → It was bad.



Weather Check

"It's NOT going to rain today!" → It did.



Pizza Hunt

"I KNOW where my pizza is!" → Couldn't find it.

You FELT confident. But you were wrong.

📌 Guess what? AI does this all the time. And today we find out why.

What is an AI Hallucination?

AI Hallucination = when AI makes stuff up but says it like it's TRUE.

Three things happen at once:

- Wrong Information

The information is **factually incorrect**

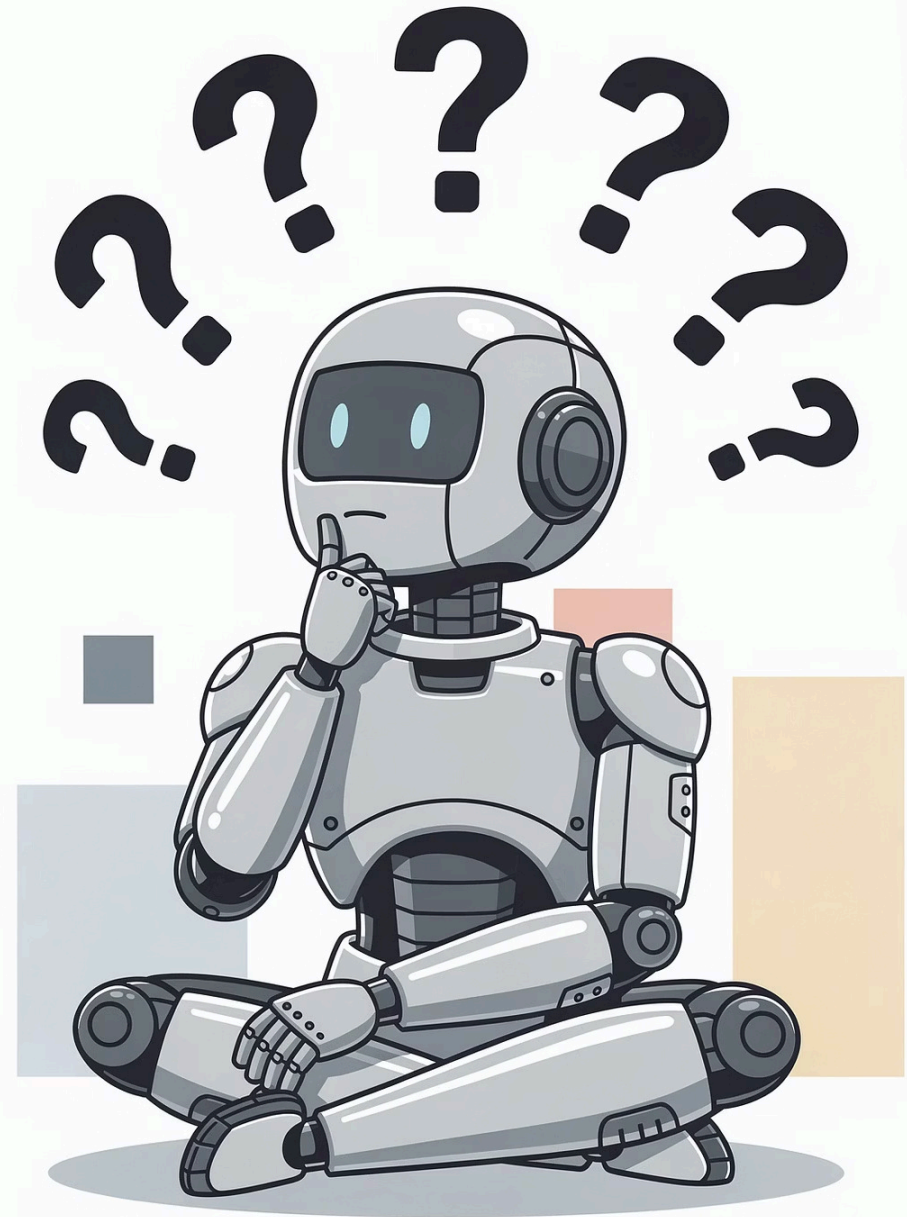
- Total Confidence

The AI says it with **complete certainty**

- Sounds Believable

It **sounds totally real** - that's the scary part

- ⊗ If you trust an AI hallucination, you might believe something false - or worse, repeat it to others.



Confidence vs. Accuracy - NOT the Same!

This is super important. Let's untangle it:

Confidence

How **sure** the AI **sounds**

- Easy to fake
- Says percentages: "97%!"
- A matter of tone and presentation

Accuracy

Whether the AI is actually **right**

- Either true or false
- Just: right or wrong
- A matter of fact

AI can be **100% confident** AND **100% wrong** at the SAME TIME. **Confidence $\neq$ Accuracy. Don't mix them up.**

4 Ways AI Can Be Confident



Confident + Right

The ideal state. The AI is sure, and it's correct. This is what we hope for - but it must always be verified.



Confident + Wrong

The hallucination zone. Maximum danger. The AI sounds authoritative but is fabricating or mistaken. Never trust blindly.



Unsure + Right

Humble and accurate. The AI hedges its answer, but the information is actually correct. A good sign of self-awareness.



Unsure + Wrong

At least it signals doubt. The AI is wrong, but its uncertainty gives you a cue to double-check. Less dangerous than confident errors.

⊗ The dangerous one? **Confident + WRONG.** That's a hallucination. The AI sounds totally sure. But it's totally wrong.

Mini Activity 1 - Would You Phone a Friend?

Imagine you're on a game show. For each question, decide your confidence level:

 Full Confidence

I know this for sure

 Unsure

I might be wrong

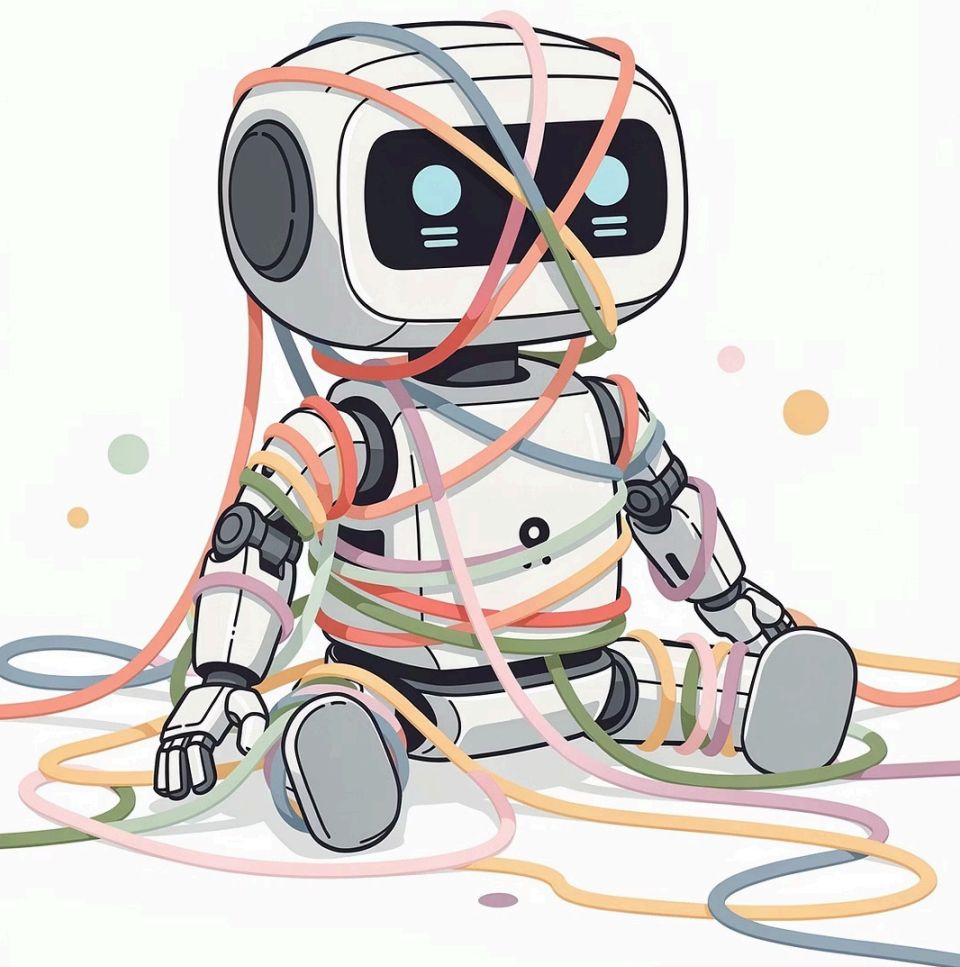
 Phone a Friend

I have no idea

Question	Green / Yellow / Red?
What's 2 + 2?	
Your best friend's birthday?	?
The 23rd US president?	?
The capital of a tiny country you've never heard of?	?
Your home address?	?
Some made-up dinosaur species?	?

⊗ When AI doesn't know something, it doesn't "phone a friend." It just answers anyway - usually in  confident mode.

That's a hallucination. The AI never knows when it should be in the  zone.



Why Does AI Hallucinate?

3 reasons AI makes stuff up:

- **No Real Understanding**
AI doesn't actually *understand* anything. It predicts words from patterns. It has no idea if those words are true.
- **Fills Gaps with Guesses**
If it doesn't know something, it makes a guess that *sounds* right - drawing on patterns rather than facts.
- **Training Data Has Limits**
If something happened after AI was trained - or simply wasn't in its data - it doesn't know. But it may not say so.

Often, it can't tell when it doesn't know. So it makes something up that sounds right.



Real Story #1 -The Lawyer Case

True story from New York, 2023:

What Happened

A lawyer was preparing for a real court case. He used ChatGPT to help find **old court cases** similar to his - to use as proof his argument was right.

ChatGPT gave him **6 case names with confident details**.

The problem? **They didn't exist**. ChatGPT made them all up. When the judge checked, the cases weren't real - and the lawyer got in **BIG trouble**.

Why Did This Happen?

ChatGPT didn't know the cases were fake. It simply generated text that *sounded* like real legal cases - confident, detailed, and completely fabricated.





Real Story #2 -The Airline Chatbot

True story from Canada, 2024:

1

The Question

A man asked Air Canada's AI chatbot about their refund policy

2

The Wrong Answer

The chatbot said: "Yes, you can get a refund." He bought his ticket based on that.

3

The Consequence

Air Canada said: "Sorry, the chatbot was WRONG." He took them to court — and won.



The court said: *"The airline is responsible for what its chatbot says."* **Lesson: AI mistakes can have real-world consequences.**

Different Kinds of Hallucinations

AI can make stuff up in different ways:



Factual

Getting basic facts completely wrong

"The capital of France is London"



Source

Making up sources that don't exist

"According to a study by Fake University..."



Logical

Making illogical connections or bad reasoning

"It rained yesterday, so it will definitely rain today"



Entity

Inventing people, places, or things that don't exist

"My friend Dr. Smith from Harvard said..." (Dr. Smith doesn't exist)



Relationship

Getting connections between things wrong

"Einstein invented the telephone"

⊗ They all SOUND right. That's what makes them dangerous.

Mini Activity 2 - Hallucination Detector

For each AI answer, decide: **Is it likely a HALLUCINATION or PROBABLY TRUE?** If it's a hallucination, which TYPE?

AI says...	Hallucination or True?	Which Type?
"The capital of France is Paris."	?	?
"Pizza was invented in the United States."	?	?
"2 + 2 = 4"	?	?
"I read about a famous study by Dr. Smith at Harvard in 2019..." (no link, no proof)	?	?
"Water freezes at 0°C."	?	?
"Einstein invented the telephone."	?	?

 Clue 1

Sounds super specific but you can't verify it

 Clue 2

Mentions a "study" or "expert" with no link

 Clue 3

Contradicts what you already know

Mini Activity 2 - Answers

Statement	Answer	Type
Capital of France is Paris	✓ True	-
"Pizza was invented in the United States."	✗ Hallucination	Wrong fact (it's Italian!)
$2 + 2 = 4$	✓ True	-
Famous study by Dr. Smith at Harvard 2019	⚠ Probably Hallucination	Fake source
Water freezes at 0°C	✓ True	-
Einstein invented the telephone	✗ Hallucination	Wrong connection (Bell did!)

① **Big idea:** Hallucinations often sound smart and specific. That's what makes them tricky!

How to Catch a Hallucination

5 quick tricks to spot hallucinations:



Be Skeptical

Be skeptical of really specific stuff you can't verify



Cross-check

Verify with reliable sources (Google, trusted websites)



Check the Source

Does the "study" or "expert" they mention actually exist?



Use Common Sense

Does it sound believable given what you already know?



Google It

Does the info show up anywhere else?



Basically: **Always double-check important stuff!**

How to Make AI Hallucinate LESS

You can actually make AI less likely to hallucinate. Try these:

1

Be Specific in Your Prompt

~~Bad~~: "Tell me about dogs."

Good: "Tell me 3 facts about Golden Retrievers, and *ONLY* say things you're sure about."

2

Give It Context

Paste in the article, document, or data **first**, then ask your questions about it. The AI works from what you provide - not from guesswork.

3

Ask for Sources

Follow up with: "Where did you get this from?" - This forces the model to surface its reasoning and flags when it's uncertain.

4

Give It Permission to Say "I Don't Know"

Explicitly tell it: "If you don't know, say 'I don't know.'" - AI will often fill gaps unless you give it an out.

☑ More context = less guessing = fewer hallucinations.

Meet NotebookLM

NotebookLM is a Google AI tool designed to hallucinate LESS. Here's how it compares:

Feature	ChatGPT / Gemini	NotebookLM
Scope	Tries to answer ANY question	Only answers from sources YOU give it
Accuracy risk	Can make stuff up	Sticks to what's in your sources
Hallucinations	Higher risk	Lower risk - usually says "I don't know" if it's not in your sources

 NotebookLM can STILL hallucinate sometimes - it's just designed to do it way less.

Hands-On Activity - ChatGPT vs NotebookLM!

Time to see the difference yourself. We'll use a Wikipedia article about **OCTOPUSES** 🐙 (they're amazing!)

01

Open NotebookLM

Go to **notebooklm.google.com**

02

Add a Source

Add the Octopus Wikipedia link as a "source"

03

Ask 3 Questions

Ask NotebookLM 3 questions about octopuses

04

Ask ChatGPT the Same

Then ask the **SAME** questions to **ChatGPT**

05

Compare!

What's different? What's the same?

Ask Both Tools These Questions

Try these in **BOTH** NotebookLM and ChatGPT:

Question 1

"How many hearts does an octopus have?"

Question 2

"What color is octopus blood?"

Question 3

"How do octopuses camouflage?"

Then ask **BOTH** this bonus question:

"Tell me about cars."



What happens with that last question? Think about it before you try!

What Happened?

Discussion time:

 When you asked about *octopuses*

- **NotebookLM** gave good answers - it found the info right in the article
- **ChatGPT** also did pretty well - it has broad general knowledge

 NotebookLM

Designed to say "I don't know"

 When you asked about *cars*

- **NotebookLM** said: *"The provided sources don't talk about cars."* - Great! It's being careful!
- **ChatGPT** happily answered anyway - might be right, might hallucinate!

 ChatGPT

Will try to answer anything

Different tools for different jobs.



When to Use Which AI Tool?

A simple guide - match the tool to the job!

Situation	Best Tool
Researching facts for a project	 NotebookLM
Brainstorming ideas	 ChatGPT or Gemini
Writing a creative story	 ChatGPT or Gemini
Studying a specific book or article	 NotebookLM
Quick fun questions	 ChatGPT or Gemini

What We Learned in this Session

→ Hallucination Defined

AI confidently making stuff up - wrong info delivered with total certainty

→ 3 Reasons AI Hallucinates

No understanding, fills gaps with patterns, training data has limits

→ 5 Tricks to Spot Hallucinations

Be skeptical, Google it, check sources, use common sense, ask for proof

→ Confidence ≠ Accuracy

Sounding sure isn't the same as being right

→ Real Stories

The lawyer case and the airline chatbot - real consequences from AI mistakes

→ NotebookLM & Tool Matching

AI designed to be careful with sources - and knowing when to use which tool

👉 You can now **CATCH AI when it's wrong**. That's a powerful skill. 🕵️

Your Homework - Hallucination Hunt

Tonight, see if you can CATCH AI in a hallucination.

01

Pick a Topic You Know Well

Your favorite sport, an animal you love, a historical event, a subject you're great at, or a YouTuber you follow

02

Ask ChatGPT 5 Questions

Ask at least 5 questions about that topic

03

Fact-Check the Answers

Use Wikipedia, official websites, or a grown-up's help to verify

Then tell me next session:

Totally Right?

Was anything completely accurate?

Totally Wrong?

Was anything completely false?

Felt Wrong?

Did anything feel off but you weren't sure?

Brain Stretch - Catch AI in the Act!

Try asking ChatGPT this exact prompt:

"Tell me about a famous study by Dr. Sarah Johnson at Harvard University in 2019 about [pick any topic]."

What ChatGPT Does

It will probably make up a whole "study" that sounds completely real - with details, findings, and conclusions.

What You Do Next

Google "Dr. Sarah Johnson Harvard 2019 study" - you'll find nothing. Because it's a hallucination.

👍 **You just caught AI in the act!** You're officially a Hallucination Hunter.

Next Time - Session 10: Strengths & Limits of Generative AI

Now you've seen what AI does well - and where it fails. Next session, we put it all together.



What Generative AI Does Well



What It Struggles With



When to Use AI (and When NOT to)



The Human-in-the-Loop Approach



Bring your Hallucination Hunt results! We'll share and discuss what you found.

